

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 21:09

File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	qwen3:8b

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 83%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.5	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	4.5	Excellent	■■■■■
Results & Evidence	4.2	Good	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	3.5	Good	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	3.5	Good	■■■■■
Ethics & Transparency	1.5	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.5 / 5.0 | Confidence: 95%

Strengths:

- [TITLE] The title is specific, informative, and includes key terms for discoverability, such as 'Physics-Informed', 'Evolutionary Artificial Neural Networks', and 'Multi-Objective Optimization'.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions. It quantifies results with statistical measures and avoids vague language.
- [KEYWORDS] The keywords are diverse, include domain-specific terms, and do not duplicate title words. They enhance discoverability and align with the content.

Weaknesses:

- [TITLE] The title is slightly long and could be shortened for clarity without losing essential information.
- [ABSTRACT] The abstract could be more concise in describing the experimental setup and results, which may improve readability.
- [KEYWORDS] The keyword 'Seismic Engineering' is somewhat broad and could be replaced with more specific terms related to structural identification.

Major Comments:

- [TITLE] Consider shortening the title to meet IEEE guidelines for brevity while retaining key technical terms.
- [ABSTRACT] The abstract could benefit from a more streamlined description of the experimental setup and results to enhance clarity and focus.
- [KEYWORDS] Replace 'Seismic Engineering' with a more specific term such as 'Seismic Response Analysis' to better align with the content.

Minor Comments:

- [TITLE] The title is technically sound and informative, though slightly verbose.
- [ABSTRACT] The abstract is well-structured and provides sufficient detail for a standalone reading.
- [KEYWORDS] The keywords are comprehensive and relevant, though some could be more specific.

Recommendations:

- [TITLE] Shorten the title to improve readability while retaining technical specificity.
- [ABSTRACT] Revise the abstract to streamline the description of the experimental setup and results.
- [ABSTRACT] Replace 'Seismic Engineering' with a more specific term related to structural identification.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear title and keywords that align with the content
- Experimental validation is present and described
- Reproducibility is emphasized through the description of the framework

Weaknesses:

- Missing literature review section which is critical for establishing context and novelty
- Lack of explicit hypothesis or research question in the introduction
- Section_vi and section_vii are duplicated content, indicating poor organization
- Section transitions are unclear and the logical flow is disrupted

Major Comments:

- The manuscript is missing a literature review section, which is essential for contextualizing the research and demonstrating novelty relative to prior work.
- The introduction does not clearly state the research problem or hypothesis, which is a critical component of a well-structured scientific paper.
- Sections VI and VII are duplicated, which indicates a significant structural flaw and poor organization.
- The logical flow between sections is inconsistent and lacks clear transitions, which undermines the coherence of the manuscript.

Minor Comments:

- The title and keywords are coherent with the body, but could be more specific to reflect the multi-objective optimization aspect.
- The abstract is missing, which is a critical component for a scientific manuscript.
- Equations are not numbered or defined, which is a major issue for IEEE standards.
- The methodology section is incomplete and lacks sufficient detail for reproducibility.

Recommendations:

- Include a comprehensive literature review section to contextualize the research and highlight novelty.
- Clearly state the research problem, hypothesis, and objectives in the introduction.
- Remove duplicated sections (VI and VII) and reorganize the content for logical flow.
- Ensure all equations are numbered and defined, and provide sufficient detail for reproducibility.
- Complete the abstract to summarize the key contributions and methodology.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 90%

Strengths:

- The paper presents a novel physics-informed EANN framework for structural parameter identification under multi-objective optimization, which is a significant contribution to the field.
- The experimental validation is thorough, with multiple statistical tests and comparisons against GA and PSO baselines.
- The paper addresses the practical challenge of acceleration-only identification, which is relevant to real-world structural health monitoring applications.
- The integration of physical constraints into the neural mapping is a key innovation that enhances the model's interpretability and robustness.

Weaknesses:

- The methodology section lacks a clear, structured presentation of the algorithm, which could hinder reproducibility.
- The transition from the theoretical formulation to the experimental validation is not clearly justified, which may reduce the paper's clarity.
- Some equations are introduced without sufficient context or explanation, which could limit their utility for independent implementation.
- The paper does not provide a detailed description of the neural network architecture or the evolutionary search strategy, which are critical for reproducibility.

Major Comments:

- The methodology section should be restructured to clearly present the algorithm and its components, including the evolutionary search strategy and the integration of physical constraints.
- The paper should provide a more detailed description of the neural network architecture and the evolutionary search strategy to ensure reproducibility.
- The transition from the theoretical formulation to the experimental validation should be more clearly justified to enhance the paper's logical flow.

Minor Comments:

- Some equations are introduced without sufficient context or explanation, which could reduce their interpretability.
- The paper could benefit from a more detailed description of the experimental setup, including the exact configuration of the five-story building prototype.
- The paper should clarify the exact implementation of the evolutionary search strategy and the multi-objective optimization formulation.

Recommendations:

- Reorganize the methodology section to present the algorithm and its components in a clear, structured manner.
- Provide a detailed description of the neural network architecture and the evolutionary search strategy to ensure reproducibility.
- Clarify the transition from the theoretical formulation to the experimental validation to enhance the paper's logical flow.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 90%

Strengths:

- Use of nonparametric tests (Kruskal-Wallis and Wilcoxon) is appropriate given the non-normal distribution of the data.
- Bootstrap confidence intervals are used to assess variability across 30 independent runs, which is a strength for reproducibility.
- The coefficient of variation (CV) is used to assess parametric robustness, which is relevant for control applications.
- RMSE, MAE, and R^2 are reported and interpreted as primary performance metrics.
- The manuscript acknowledges the need for physical consistency in parameter estimation, which is critical for control-oriented modeling.

Weaknesses:

- Bootstrap confidence intervals are not described in terms of resample size or variance, which could affect the interpretation of results.
- The Holm correction is not explicitly detailed in the methodology, which could affect the reproducibility of the statistical analysis.
- The coefficient of variation (CV) is reported but not interpreted in the context of practical significance for control applications.
- The hypervolume indicator is used but not compared to state-of-the-art baselines, which limits the novelty of the approach.
- The manuscript does not provide a detailed power analysis to justify the sample size of 30 runs.

Major Comments:

- The manuscript should provide more detailed information on the bootstrap confidence intervals, including the number of resamples and variance of the estimates.
- The Holm correction should be explicitly described in the methodology section to ensure reproducibility.
- The coefficient of variation (CV) should be interpreted in the context of practical significance for control applications.
- The hypervolume indicator should be compared to state-of-the-art baselines to demonstrate the novelty of the approach.

Minor Comments:

- The manuscript should clarify the number of resamples used in bootstrap confidence intervals.
- The manuscript should provide more details on the power analysis to justify the sample size of 30 runs.
- The manuscript should explicitly describe the Holm correction in the methodology section.

Recommendations:

- Include a detailed description of the bootstrap confidence intervals, including the number of resamples and variance of the estimates.
- Provide a power analysis to justify the sample size of 30 runs.
- Clarify the Holm correction in the methodology section to ensure reproducibility.
- Interpret the coefficient of variation (CV) in the context of practical significance for control applications.
- Compare the hypervolume indicator to state-of-the-art baselines to demonstrate the novelty of the approach.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 75%

Strengths:

- The manuscript presents a clear and detailed formulation of the structural modeling and identification problem, with well-defined equations and mathematical rigor.
- The experimental setup is described with sufficient detail to support reproducibility, including the use of specific sensors and their calibration.
- The comparison of algorithms (GA, PSO, EANN) is well-structured, with clear performance metrics and statistical validation.

Weaknesses:

- Several figures and tables are referenced but their content is not visible, which limits the ability to verify their coherence with the text and the claims made.
- Equations (11), (13), and (43) are referenced but their full content is not visible, which could affect the verification of mathematical correctness and coherence.
- The text refers to specific numerical values (e.g., 0.021, 0.023, 11.7, 11.8) but these are not clearly tied to any table or figure, raising concerns about numerical consistency.

Major Comments:

- The manuscript lacks visibility of key figures, tables, and equations, which are critical for verifying the claims made in the text and ensuring reproducibility. This is a major issue for IEEE publication standards.

Minor Comments:

- Some figures and tables are referenced without sufficient detail in their captions, which could hinder the reader's ability to understand the content without reading the full paper.

Recommendations:

- Please ensure that all figures, tables, and equations are fully visible and self-explanatory, with clear captions and references in the text.
- Verify that all equations are fully presented and that their variables are clearly defined before or immediately after their use.
- Ensure that all numerical values in the text are explicitly tied to tables or figures for consistency and clarity.

ResultsReviewer

Results validity and empirical support

Score: **4.2 / 5.0** | Confidence: 80%

Strengths:

- The results clearly address the stated objectives of the study, including the comparison of the EANN framework against GA and PSO baselines.
- Quantitative metrics such as MSE, RMSE, CV, and hypervolume are reported, with sufficient detail to support the claims.

- The results are consistent across the text, figures, and tables, with clear statistical validation through Kruskal-Wallis and Wilcoxon rank-sum tests.
- The manuscript provides a comprehensive comparison with state-of-the-art baselines, including GA and PSO, and reports performance improvements.

Weaknesses:

- Negative results or limitations of the proposed framework are not explicitly reported, which could affect the perceived robustness and generalizability of the approach.
- The manuscript does not provide detailed descriptions of the specific metrics used for the Pareto front analysis, such as the exact calculation of hypervolume or spread indicators.
- The exact values of the statistical tests (e.g., p-values, confidence intervals) are not provided, which limits the ability to fully assess the significance of the results.
- The manuscript does not clarify whether the reported MSE and RMSE values are per-floor or aggregated across all floors, which could affect the interpretation of the results.

Major Comments:

- The manuscript should explicitly report any negative results or limitations of the EANN framework to provide a balanced view of its performance.

Minor Comments:

- The exact calculation of the hypervolume and spread indicators should be clarified in the results section to ensure reproducibility and transparency.
- The manuscript should specify whether the reported MSE and RMSE values are per-floor or aggregated across all floors to avoid misinterpretation.

Recommendations:

- Include a section on limitations and negative results to enhance the credibility and completeness of the study.
- Clarify the exact calculation and reporting of hypervolume and spread indicators in the Pareto front analysis.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- The Discussion section effectively interprets the results in the context of structural parameter identification under seismic conditions, highlighting the statistical significance of performance differences.
- The Conclusions section clearly states the contribution of the EANN framework to the field of structural health monitoring and adaptive control.
- The manuscript provides a clear experimental validation against GA and PSO baselines, which is a strong point for reproducibility and technical rigor.

Weaknesses:

- The Discussion section lacks a thorough comparison with prior work in the literature, which is critical for establishing the novelty and relevance of the EANN framework.
- The Discussion does not address unexpected findings or limitations of the study, such as the higher computational cost of EANN compared to GA and PSO.
- The Conclusions section does not specify concrete future work directions, which is essential for guiding further research in the field.
- The Discussion section overgeneralizes the implications of the results, particularly in the context of real-world applications without sufficient justification.

Major Comments:

- The Discussion section should include a more detailed comparison with prior IEEE publications to better establish the novelty and technical contribution of the EANN framework.

- The manuscript should explicitly address the limitations of the study, such as the computational cost of EANN, to provide a balanced view of the proposed method.

Minor Comments:

- The Discussion section could benefit from a more precise explanation of the statistical significance and its practical implications for control deployment.
- The Conclusions section should be more specific in quantifying the performance improvements of EANN over GA and PSO.

Recommendations:

- Include a detailed comparison with prior IEEE publications in the Discussion section to highlight the novelty and technical contribution of the EANN framework.
- Clarify the limitations of the study, particularly the computational cost of EANN, and discuss how these limitations might affect real-world applications.
- Specify concrete future work directions, such as exploring hybrid models or optimizing the EANN framework for lower computational cost.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical contribution and experimental validation
- Well-defined mathematical framework
- Robust statistical validation methods
- Reproducibility emphasized through detailed experimental setup

Weaknesses:

- Overuse of passive voice
- Vague quantifiers without numerical support
- Unexplained acronyms
- Lack of precise algorithm description for reproducibility
- Weak transitions between sections

Major Comments:

- The manuscript lacks precise algorithm description for reproducibility. The EANN framework should be described with sufficient detail for independent implementation.
- The use of passive voice is overused and should be minimized for clarity and directness.
- The acronyms EANN, PSO, and GA are not defined on first use and should be spelled out in the abstract and introduction.

Minor Comments:

- The phrase 'significant' is used without numerical support and should be replaced with specific values.
- The sentence 'many' is vague and should be replaced with a specific number.
- The phrase 'several' is vague and should be replaced with a specific number.

Recommendations:

- Define all acronyms on first use, including EANN, PSO, and GA.
 - Revise passive voice constructions to active voice where possible.
 - Replace vague quantifiers with specific numerical values to enhance clarity and precision.
 - Provide a detailed algorithm description for the EANN framework to ensure reproducibility.
 - Improve transitions between sections to enhance logical flow.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- A good mix of recent (2021-2023) and classic references
- Citations are consistent with IEEE style
- References directly support the claims made in the introduction and methodology

Weaknesses:

- Lack of seminal foundational papers on structural health monitoring and vibration control theory
- Missing key references on state estimation and observer design
- Limited diversity in authors and institutions

Major Comments:

- The references section lacks foundational seminal works in structural health monitoring and vibration control, which are critical for establishing the context and novelty of the work.

Minor Comments:

- The book by Chopra [29] is relevant but should be supplemented with seminal papers in the field for completeness.
- The references section could benefit from more diversity in authors and institutions to reflect a broader academic landscape.

Recommendations:

- Include seminal papers on structural health monitoring and vibration control theory, such as those by Clough and Penzien (1975) or those by Brincker et al. (1993).
- Add key references on state estimation and observer design, such as those by Luenberger (1964) or Kalman (1960).
- Ensure that the references section includes a broader range of authors and institutions to reflect a more diverse academic landscape.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The methodology is well-defined and focuses on a novel approach to structural parameter identification using a physics-informed EANN framework.
- The experimental validation is presented with figures and metrics, which supports the claims made in the paper.
- The paper addresses a significant limitation of classical metaheuristic approaches by proposing a decoupled training and deployment paradigm.

Weaknesses:

- The manuscript lacks a comprehensive discussion of limitations, which is critical for a Q1 journal publication.
- The limitations are not specific or detailed, which undermines the credibility of the conclusions.
- The reproducibility of the results is not addressed, as there is no mention of data or code availability.
- The manuscript does not provide a clear statement of author contributions, which is a standard requirement for transparency.

Major Comments:

- The manuscript fails to meet the ethical reporting standards required by IEEE and Q1 journals. Ethical approval, data availability, conflict of interest, and funding disclosure are all missing.
- The limitations section is entirely absent, which is a major shortcoming for a high-impact journal. The paper should include a detailed discussion of the limitations and their impact on the conclusions.

- Reproducibility is not addressed, which is a critical requirement for IEEE publications. The manuscript should provide information on data and code availability to enable independent verification.

Minor Comments:

- The title, abstract, and keywords are coherent with the body of the manuscript, but they lack the necessary detail to fully reflect the scope of the research.
- The references are not provided, which makes it difficult to assess the novelty and relevance of the work in the context of prior IEEE publications.

Recommendations:

- Include a section on ethical approval, data availability, conflict of interest, and funding sources.
- Add a detailed limitations section that explicitly states the limitations, their impact on the conclusions, and how they affect the generalizability of the findings.
- Provide a reproducibility statement that includes information on data and code availability to enable independent implementation and validation.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [TITLE] Consider shortening the title to meet IEEE guidelines for brevity while retaining key technical terms.
- [2] [ABSTRACT] The abstract could benefit from a more streamlined description of the experimental setup and results to enhance clarity and focus.
- [3] [KEYWORDS] Replace 'Seismic Engineering' with a more specific term such as 'Seismic Response Analysis' to better align with the content.
- [4] The manuscript is missing a literature review section, which is essential for contextualizing the research and demonstrating novelty relative to prior work.
- [5] The introduction does not clearly state the research problem or hypothesis, which is a critical component of a well-structured scientific paper.
- [6] Sections VI and VII are duplicated, which indicates a significant structural flaw and poor organization.
- [7] The logical flow between sections is inconsistent and lacks clear transitions, which undermines the coherence of the manuscript.
- [8] The methodology section should be restructured to clearly present the algorithm and its components, including the evolutionary search strategy and the integration of physical constraints.
- [9] The paper should provide a more detailed description of the neural network architecture and the evolutionary search strategy to ensure reproducibility.
- [10] The transition from the theoretical formulation to the experimental validation should be more clearly justified to enhance the paper's logical flow.
- [11] The manuscript should provide more detailed information on the bootstrap confidence intervals, including the number of resamples and variance of the estimates.
- [12] The Holm correction should be explicitly described in the methodology section to ensure reproducibility.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] The title is technically sound and informative, though slightly verbose.
- [2] [ABSTRACT] The abstract is well-structured and provides sufficient detail for a standalone reading.
- [3] [KEYWORDS] The keywords are comprehensive and relevant, though some could be more specific.
- [4] The title and keywords are coherent with the body, but could be more specific to reflect the multi-objective optimization aspect.
- [5] The abstract is missing, which is a critical component for a scientific manuscript.
- [6] Equations are not numbered or defined, which is a major issue for IEEE standards.
- [7] The methodology section is incomplete and lacks sufficient detail for reproducibility.
- [8] Some equations are introduced without sufficient context or explanation, which could reduce their interpretability.
- [9] The paper could benefit from a more detailed description of the experimental setup, including the exact configuration of the five-story building prototype.
- [10] The paper should clarify the exact implementation of the evolutionary search strategy and the multi-objective optimization formulation.
- [11] The manuscript should clarify the number of resamples used in bootstrap confidence intervals.
- [12] The manuscript should provide more details on the power analysis to justify the sample size of 30 runs.

Specific Improvement Recommendations

1. [TITLE] Shorten the title to improve readability while retaining technical specificity.

2. [ABSTRACT] Revise the abstract to streamline the description of the experimental setup and results.
3. [ABSTRACT] Replace 'Seismic Engineering' with a more specific term related to structural identification.
4. Include a comprehensive literature review section to contextualize the research and highlight novelty.
5. Clearly state the research problem, hypothesis, and objectives in the introduction.
6. Remove duplicated sections (VI and VII) and reorganize the content for logical flow.
7. Ensure all equations are numbered and defined, and provide sufficient detail for reproducibility.
8. Complete the abstract to summarize the key contributions and methodology.
9. Reorganize the methodology section to present the algorithm and its components in a clear, structured manner.
10. Provide a detailed description of the neural network architecture and the evolutionary search strategy to ensure reproducibility.
11. Clarify the transition from the theoretical formulation to the experimental validation to enhance the paper's logical flow.
12. Include a detailed description of the bootstrap confidence intervals, including the number of resamples and variance of the estimates.

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.